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Lab Project Name: Vehicle Parking Management System

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[For Teachers use only: **Don't Write Anything inside this box**]

Lab Project Status

Marks:

Signature:

Comments:

Date:

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Chapter 1

Introduction

1.1 Introduction

Parking management system for managing the records of the incoming and outgoing vehicles in a parking house. It's an easy for Admin to retrieve the data if the vehicle has been visited through number, he can get that data

Now days in many public places such as malls, multiplex system, hospitals, offices, market areas there is a crucial problem of vehicle parking. The vehicle parking area has many lanes/slots for car parking. So, to park a vehicle one has to look for all the lanes. Moreover, this involves a lot of manual labors and investment. Instead of vehicle caught in towing the vehicle can park on safe and security with low cost. Parking control system has been generated in such a way that it is filled with many secure devices such as, parking control gates, toll gates, time and attendance machine, car counting system etc. These features are hereby very necessary nowadays to secure your car and also to evaluate the fee structure for every vehicle entry and exit the objective of this project is to build a Vehicle Parking management system that enables the time management and control of vehicles using number plate recognition. The system that will track the entry and exit of cars, maintain a listing of cars within the parking lot, and determine if the parking lot is full or not. It will determine the cost of per vehicle according to their time consumption.

1.2 Objective

We can park our vehicle in our own slot by paying.

- Because of that there is no towing problems.
- And our vehicle has been parked as a secure condition.
- There is no risk for vehicle owner for parking the car.
- In case of any damages and problem of vehicle that will claim by parking management.
- As the world is facing many threads daily, robberies are done easily with no track to trace, bomb blasts occur with the use of vehicle, so if a proper system is adopted each and every record can be saved and anyone can be track easily therefore mainly is to make a better and fast software, most important user-friendly
- Maintain records in short time of period.
- Determines the parking area is full or not.
- Enhances the visitor's experience

Definition of problem:

- Now a days in parking like valet parking they maintain just with the tokens and they have records the vehicle details in books so that during some critical situations like police enquiry of terrorist car or vehicle referrer that case it is difficult to find the details of particular vehicle but in this case is easy to find in 1to 2 seconds
- By parking the vehicle in public place the vehicle can be claimed by towing person but in this case there is no towing problems and no need to give fine for anything we can park our vehicle with securely

Vehicle Parking Details

This report is very essential in this system. This report provides a brief summary of vehicle activities. It shows the overall Entry and Exit time. It shows the User at time of Entry and Exit .It also provides the facility for examining the total vehicle details according to date wise.

Transaction Details

This report will show the Transaction between the customer and the System. . It shows the cost of the vehicle after using the facility of parking. It will show the number of transactions by date wise. It will also have User at time of the Transaction.

Hardware and software requirement

PROCESSOR TYP: Pentium IV or above for optimum performance.

SYSTEM RAM: 1.00GB and Above

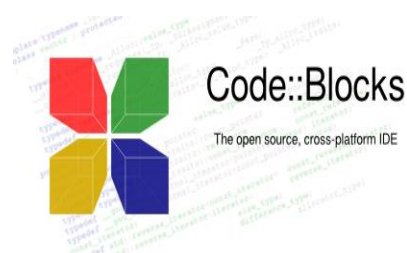
INPUT DEVICE: BASIC KEYBOARD AND TOUCH PAD

OUTPUT DEVICE: STANDARD COLOR MONITOR

OPERATING SYSTEM: WINDOWS7,8,9,10,11

FRONT END: Code Blocks

BACK END: 'C' Program



Chapter 2

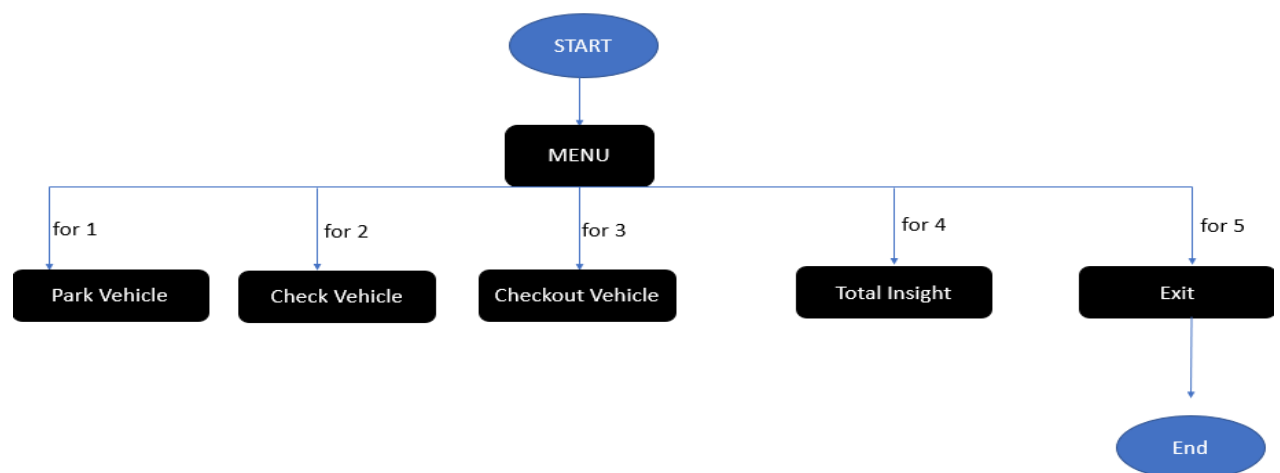
Design/Development/Implementation of the Project

2.1 Main Function

Algorithm

1. Take choice value from user.
2. Use switch case for choice base menu . Insert Main menu by using case(like 1,2,3) .
3. In Case 1 call insert function.
4. In case 2 call check vehicle function.
5. In case 3 call checkout function
6. In case 4 call Insight function
7. In case 5 call exit function

Diagram of Main function

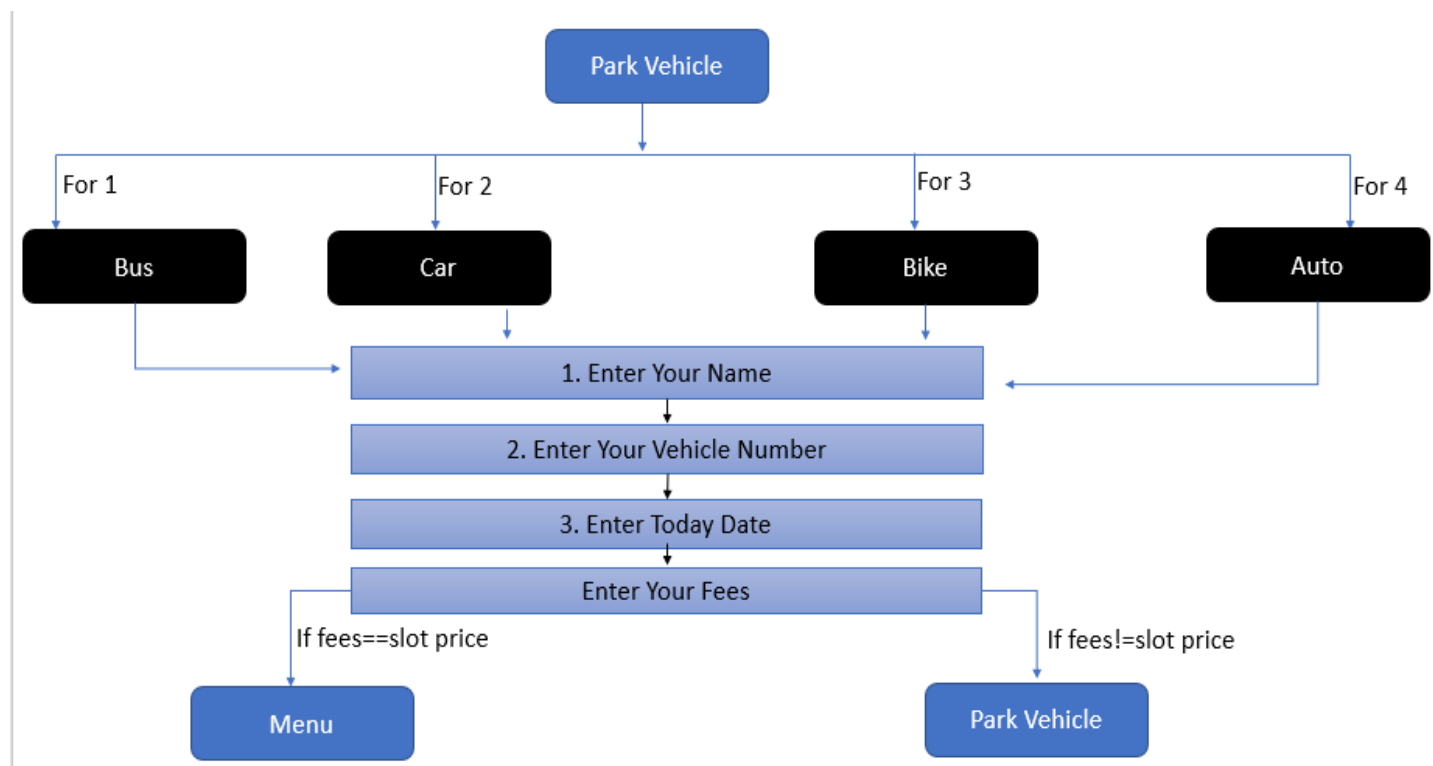


2.2 Insert Function

Algorithm

1. Take Name from user and store it in a variable.
2. Take Vehicle number from user and input it.
3. Take Today date from user and store it.
4. Now take the parking fixed fees from user if user give less or grater than fixed amount then provide other chance for user.
5. Then call Receipt function.
6. And decrease slot number, increase index number.

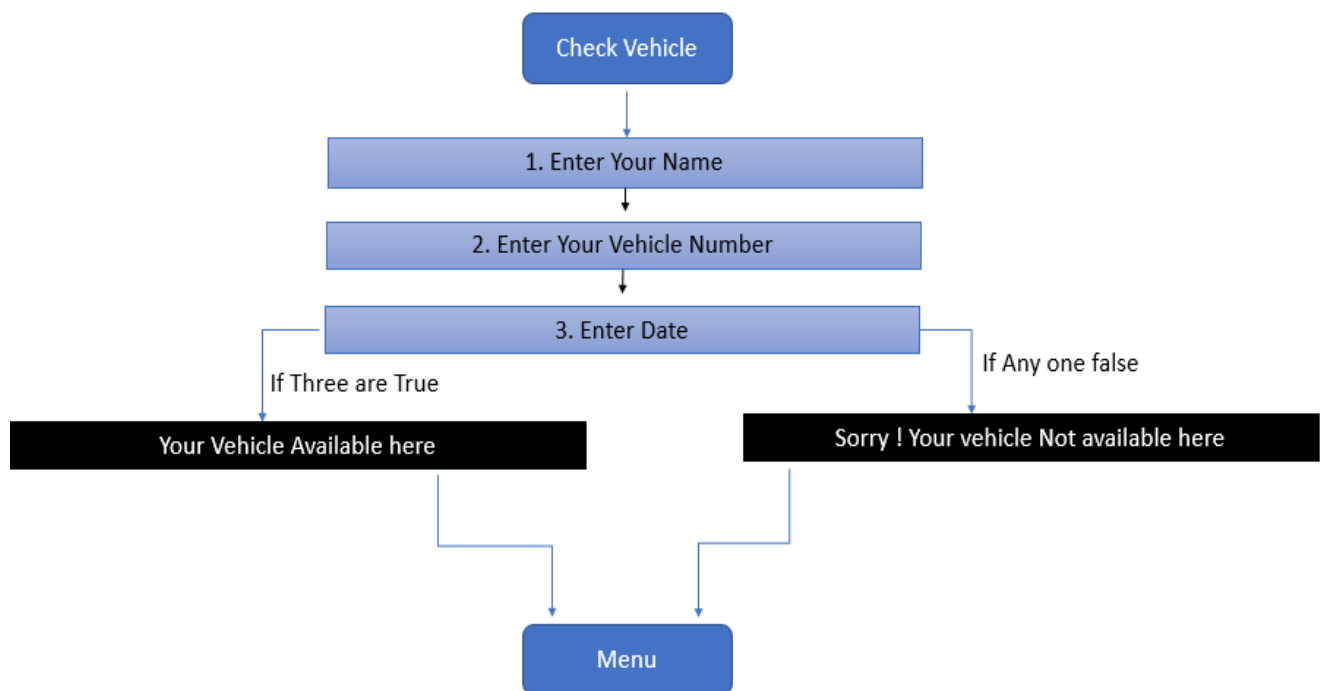
Diagram Of Insert Function:



2.3 Check Vehicle Function

1. Take use name from user.
2. Take vehicle number from user.
3. Take date from user.
4. Then compare all data take from user to stored data.
5. If all is return true then print your vehicle is here.
6. If any one is return false then print your vehicle not available here.

Diagram Of check vehicle Function



2.4 Checkout Vehicle Function

Algorithm

1. Take value from user and store it in a variable.
2. Now build a mini user manual what kind of vehicle user need checkout.
3. Build a choice base menu using switch case and use previous value which take from user
4. Case 1 call checkout bus.
5. Case 2 call checkout car.
6. Case 3 call checkout bike.
7. Case 4 call checkout auto.
8. If any other option get another chance .

2.4.1 Checkout Bus

Algorithm

1. Take a index number from user.
2. If index number find in database then delete the index all value from the database
3. Increase the bus slot.
4. Print vehicle checkout done.

2.4.2 Checkout Car

Algorithm

1. Take a index number from user.
2. If index number find in database then delete the index all value from the database
3. Increase the car slot.
4. Print vehicle checkout done.

2.4.3 Checkout Bike

Algorithm

1. Take a index number from user.
2. If index number find in database then delete the index all value from the database
3. Increase the Bike slot.
4. Print vehicle checkout done.

2.4.4 Checkout Auto

Algorithm

1. Take a index number from user.
2. If index number find in database then delete the index all value from the database
3. Increase the Auto slot.
4. Print vehicle checkout done.

Chapter 3

3.1 Implementation

```
void main()
{
    first();

    int choose;
    do
    {
        second();
again:
        printf("\nPlease choose any option: ");
        scanf("%d", &choose);
        switch (choose)
        {
            case 1:
                park_vehicle();
                break;
            case 2:
                check_vehicle();
                break;
            case 3:
                check_out();
            case 2:
                check_vehicle();
                break;
            case 3:
                check_out();
            case 4:
                check_insight();
                break;
            case 5:
                printf("Thank you for using our parking stand\n");
                exit(0);
                break;
            default:
                printf("Invalid number try again\n");
                goto again;
                break;
        }
    }
    while (choose != 5);
    return 0;
}
```

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
int bus_slot = 5, car_slot = 5, bike_slot = 20, auto_slot = 15;
int earings = 0, bus_count = 0, car_count = 0, bike_count = 0, auto_count = 0;
int index1 = 0, index2 = 0, index3 = 0;
struct customer
{
    char customerName[10][20], date[10][13];
    int vehicleNumber[10];
} data;
void print_receipt(int fees)
{
    printf("\n\t\t\t\t\t\t\t\t\t\t Here you receipt!\n");
    printf("\t\t\t\t\t\t\t\t\t\t*****\n");
    printf("\t\t\t\t\t\t\t\t\t\t Name: %s\n", data.customerName[index2]);
    printf("\t\t\t\t\t\t\t\t\t\t Vehicle numebr: %d\n", data.vehicleNumber[index1]);
    printf("\t\t\t\t\t\t\t\t\t\t Date: %s\n", data.date[index3]);
    printf("\t\t\t\t\t\t\t\t\t\t Paid Ammount : %d\n", fees);
    printf("\t\t\t\t\t\t\t\t\t\t*****\n");
}
void bus_park()
{

```

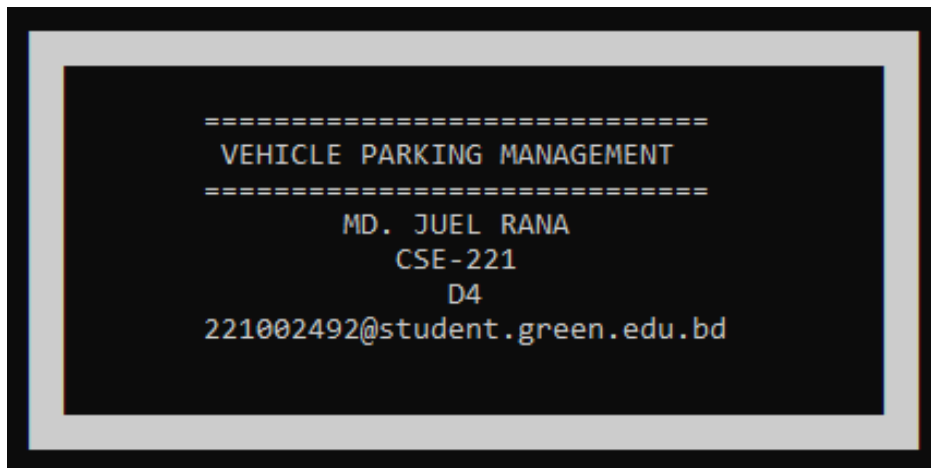
```

char customerName1[20], date1[13];
int vehicleNumber,temp,j,k,low=0,mid,high=index1;
int i = 0, check1, check2=1, check3;
printf("\nPlease enter some details\n");
printf("Enter Name: ");
fflush(stdin);
gets(customerName1);
printf("Enter vehicle number: ");
scanf("%d", &vehicleNumber);
printf("Enter date: ");
fflush(stdin);
gets(date1);
for(k= 0; k<=index1; k++)
{
    for(j= 0; j<=index1-k-1; j++)
    {
        if(data.vehicleNumber[j]>data.vehicleNumber[j+1])
        {
            temp= data.vehicleNumber[j];
            data.vehicleNumber[j]=data.vehicleNumber[j+1];
            data.vehicleNumber[j+1]=temp;
        }
    }
}

```

3.2 Results and Discussions

3.2.1 Results



```
Please choose any option: 1

1.Bus                      2.Car
3.Bike                    4.Auto
Which vehicle you want to park: 2

Your Parking Slot Number : 1
*****
Enter your name: JUEL
Enter vehicle number: 221
Enter today's date: 1-1-2023
Enter parking fees: 150
Your payment is successfully done
```

```
Please choose any option: 2

Please enter some details
Enter Name: JUEL
Enter vehicle number: 221
Enter date: 1-1-2023

Your vehicle is available here
```

Chapter 4

Conclusion

4.1 Introduction

This Project is minimizing the task of parking a vehicle by paying and saying some details about customer and vehicle to save data. In this the vehicle is parked as a safe and secure. This project is done as Efficient as possible Hereby I, the Student of BSc (CSE) 3rd Semester concludes that the project was completely and slowly developed by me. I also conclude that this project has helped us gain more knowledge about the topic that we are indulged ourselves into “Code Blocks”. I would be glad to enhance and promote this project if given chance and help ourselves and society in the near future. The developed application is tested with sample inputs and outputs obtained in according to the requirement. Even though I have tried our level best to make it a dream project. Due to time constraints, I could not add more facilities to it. The efficiency of the developed system can be enhanced with some minor modifications. Future development can be made in proposed system by integration more services like

4.1 Practical Implications

I am trying my best in this project. In future the project would be better. using More function and data structure more concept that make this project perfect. If Project space complexity and time complexity are less than previous then it will be perfect. The efficiency of the developed system can be enhanced with some minor modifications. Future development can be made in proposed system by integration more services like:

1. It can be implemented through web pages.
2. New effectives modules can be added time to time.

4.2 Scope of Future Work

In the modern age. Many people have vehicles. Vehicle is now a basic need. Everyplace is under the process of urbanization. There are many corporate offices and shopping centers etc. There are many recreational places where people used to go for refreshment. So, all these places need a parking space where people can park their vehicles safely and easily. Every parking area needs a system that records the detail of vehicles to give the facility. These systems might be computerized or non-computerized. With the help of computerized system, we can deliver a good service to customer who wants to park their vehicle into the any organization's premises.

Vehicle parking management system is an automatic system which delivers data processing in very high speed in systematic manner. Parking is a growing need of the time. Development of this system is very useful in this area of field. We can sell this system to any organization. By using our system, they can maintain records very easily. Our system covers every area of parking management. In coming future there will be excessive need of Vehicle parking management system.

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